

Fig. 3: HF (13.56MHz) tag example

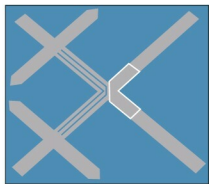


Fig. 4: UHF (860-930MHz) tag example

Having just a simple code of up to 256 bits would lead to smaller chip size and hence lower tag costs, which is recognised as the key factor for widespread adoption of RFID in the supply chain. Tags that store just an ID number are often called licence plate tags.

Tag classes

One of the main ways of categorising RFID tags is by their capability to read and write data. This leads to the following four classes:

Class 0 (read-only, factory-programmed). These are the simplest type of tags, where the data, which is usually a simple ID number (EPC), is written only once into the tag during manufacture. The memory is then disabled from any further updates. Class 0 is also used to define a category of tags called electronic article surveillance or anti-theft devices, which have no ID and announce their presence only when passing through an antenna field.

Class 1 (write-once read-only, factory- or user-programmed). In this case,

TABLE III
Different Tag Classes

Class	Known as	Memory	Power source	Application
0	EAS, EPC	None, EPC	Passive	Anti-theft, ID
1	EPC	Read-only	Any	Identification
2	EPC	Read-write	Any	Data logging
3	Sensor tags	Read-write	Semi-passive/active	Sensors
4	Smart dust	Read-write	Active	Ad-hoc networking

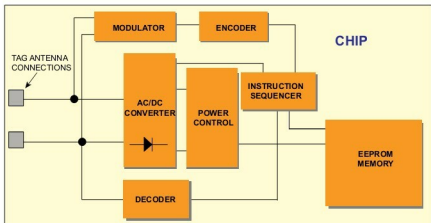


Fig. 5: Basic tag IC architecture

the tag is manufactured with no data written into the memory. Data can then either be written by the tag manufacturer or by the user one time. Following this no further writes are allowed and the tag can only be read. Tags of this type usually act as simple identifiers.

Class 2 (read-write). These are the most flexible type of tags, where users have access to read and write data into the tag's memory. They are typically used as data loggers and therefore contain larger memory space than what is needed for just a simple ID number.

Class 3 (read-write with on-board sensors). These tags contain on-board sensors for recording parameters like temperature, pressure and motion by writing into the tag's memory. As sensor readings must be taken in the absence of a reader, the tags are either semi-passive or active.

Class 4 (read-write with integrated transmitters). These are like miniature radio devices which can communicate with other tags and devices without the presence of a reader. This means that they are completely active with their own battery power source.

Selecting a tag

Choosing the right tag for a particular RFID application is an important consideration, and should take into account many of the factors listed below:

1. Size and form factor—where does the tag have to fit?
2. How close will the tags be to each other?
3. Durability—does the tag need to have a strong outer protection against regular wear and tear?
4. Is the tag reusable?
5. Resistance to harsh (corrosive, steamy, etc) environments
6. Polarisation—the tag's orientation with respect to the reader field
7. Exposure to different temperature ranges
8. Communication distance
9. Influence of materials such as metals and liquids
10. Environment (electrical noise, other radio devices and equipment)
11. Operating frequency (LF, HF or UHF)
12. Supported communication